Convexity in Several Variables

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Source

This section is adapted from the convexity part of Zhou [1], used under the MIT License (see the license text in Linear Algebra). The source states the first-order test (in one direction) and the second-derivative test for convexity without proof; here both directions of the first-order test are proved, and the second-derivative test is proved from the Taylor theorem of Section in Critical Points and Optimality Conditions. The source defines both concave and strictly concave as "$-f$ is strictly convex"; here f is concave when $-f$ is convex. The source states that an intersection of convex sets is convex and that a maximum of convex functions is convex for any finite number of them; here each is stated for two and the general case follows by induction. These parts of the source are not part of this section:

- its catalogs of convex sets and convex functions, including the positive semidefinite matrices and $-\log\det$
- intersections of infinitely many convex sets, and suprema of infinitely many convex functions
- its examples, exercises and figures, including the multivariate normal maximum likelihood example

Definition 0.1 (Convex set)**.** A set $K \subseteq \mathbb{R}^p$ is **convex** if [2] for all $\underset{\sim}{x}, \underset{\sim}{y} \in K$ and all $t \in [0,1]$, the point $t\underset{\sim}{x} + (1-t)\underset{\sim}{y}$ is in K: K contains the whole line segment between any two of its points.

Example 0.1 (Convex sets)**.**

- $\mathbb{R}^p$ is convex, and so is any set with at most one point.
- The closed ball $B = \{\underset{\sim}{x} : \|\underset{\sim}{x}\| \le r\}$ is convex: for $\underset{\sim}{x}, \underset{\sim}{y} \in B$ and $t \in [0,1]$,

$$\begin{aligned}
\|t\underset{\sim}{x} + (1-t)\underset{\sim}{y}\| &\le \|t\underset{\sim}{x}\| + \|(1-t)\underset{\sim}{y}\| && \text{(triangle inequality)}\\
&= t\,\|\underset{\sim}{x}\| + (1-t)\,\|\underset{\sim}{y}\| && \text{(norm of a multiple; } t, 1-t \ge 0)\\
&\le t\,r + (1-t)\,r && (\underset{\sim}{x}, \underset{\sim}{y} \in B)\\
&= r. && \text{(arithmetic)}
\end{aligned}$$

The first step is the triangle inequality, and the second uses the norm of a multiple.

Example 0.2 (A set that is not convex)**.** $K = \{x \in \mathbb{R} : |x| \ge 1\}$ is not convex: -1 and 1 are in K, but with $t = \frac{1}{2}$ the point $\frac{1}{2}(-1) + \frac{1}{2}(1) = 0$ is not.

Definition 0.2 (Half-space)**.** Let $\underset{\sim}{a} \in \mathbb{R}^p$ be a nonzero vector and c a real number. The set $\{\underset{\sim}{x} \in \mathbb{R}^p : \underset{\sim}{a} \cdot \underset{\sim}{x} \ge c\}$ is a (closed) **half-space**: the points on one side of the boundary $\{\underset{\sim}{x} : \underset{\sim}{a} \cdot \underset{\sim}{x} = c\}$, together with the boundary. In $\mathbb{R}^2$, a half-space is called a **half-plane**.

Example 0.3 (The right half-plane)**.** With $\underset{\sim}{a} = (1,0)^\top$ and $c = 0$, the half-plane $\{\underset{\sim}{x} \in \mathbb{R}^2 : x_1 \ge 0\}$ is the set of points on or to the right of the x_2-axis. $(1,5)^\top$ is in it, since $1 \ge 0$; $(0,-3)^\top$ is in it, on the boundary; and $(-1,0)^\top$ is not, since $-1 < 0$.

Theorem 0.1 (An intersection of convex sets is convex)**.** *If K_1 and K_2 are convex subsets of $\mathbb{R}^p$ (Definition 0.1), then $K_1 \cap K_2$ is convex.*

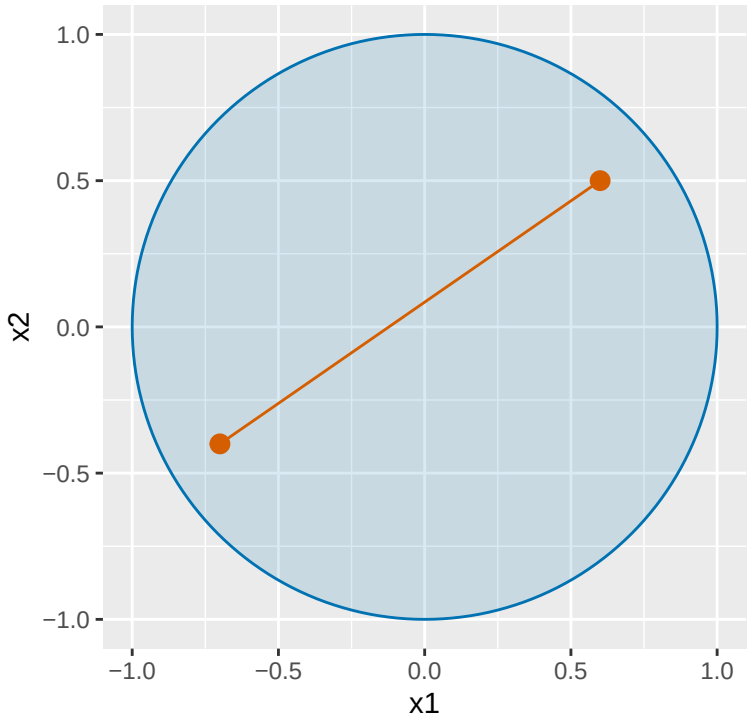
Proof

Proof. Let $\underset{\sim}{x}, \underset{\sim}{y} \in K_1 \cap K_2$ and $t \in [0,1]$. Then $\underset{\sim}{x}, \underset{\sim}{y} \in K_1$, so $t\underset{\sim}{x} + (1-t)\underset{\sim}{y} \in K_1$, since K_1 is convex; in the same way $t\underset{\sim}{x} + (1-t)\underset{\sim}{y} \in K_2$. So $t\underset{\sim}{x} + (1-t)\underset{\sim}{y} \in K_1 \cap K_2$. □

Example 0.4 (Intersecting a ball with a half-plane)**.** The closed unit ball in $\mathbb{R}^2$ is convex (Example 0.1), and so is the half-plane (Definition 0.2) $\{\underset{\sim}{x} : x_1 \ge 0\}$: if $x_1 \ge 0$ and $y_1 \ge 0$, then $tx_1 + (1-t)y_1 \ge 0$ for $t \in [0,1]$. By Theorem 0.1, the closed half-disk $\{\underset{\sim}{x} : \|\underset{\sim}{x}\| \le 1,\ x_1 \ge 0\}$ is convex. A union need not be convex: $\{x \in \mathbb{R} : x \le -1\}$ and $\{x \in \mathbb{R} : x \ge 1\}$ are convex, but their union is the set of Example 0.2.
Figure 2 shows the half-disk.

R

```r
cs_angle_fc <- seq(0, 2 * pi, length.out = 201)
cs_cos_fc <- cos(cs_angle_fc)
cs_sin_fc <- sin(cs_angle_fc)
cs_disk_fc <- tibble::tibble(x1 = cs_cos_fc, x2 = cs_sin_fc)
cs_points_x1_fc <- c(-0.7, 0.6)
cs_points_x2_fc <- c(-0.4, 0.5)
cs_points_fc <- tibble::tibble(x1 = cs_points_x1_fc, x2 = cs_points_x2_fc)
cs_segment_fc <- tibble::tibble(
  x1 = -0.7, x2 = -0.4, x1_end = 0.6, x2_end = 0.5
)
cs_aes_fc <- ggplot2::aes(x1, x2)
cs_seg_aes_fc <- ggplot2::aes(xend = x1_end, yend = x2_end)
ggplot2::ggplot(cs_disk_fc, cs_aes_fc) +
  ggplot2::geom_polygon(fill = "#0072B2", alpha = 0.15, colour = "#0072B2") +
  ggplot2::geom_segment(
    data = cs_segment_fc, mapping = cs_seg_aes_fc, colour = "#D55E00"
  ) +
  ggplot2::geom_point(data = cs_points_fc, colour = "#D55E00", size = 3) +
  ggplot2::coord_fixed() +
  ggplot2::labs(x = "x1", y = "x2")
```



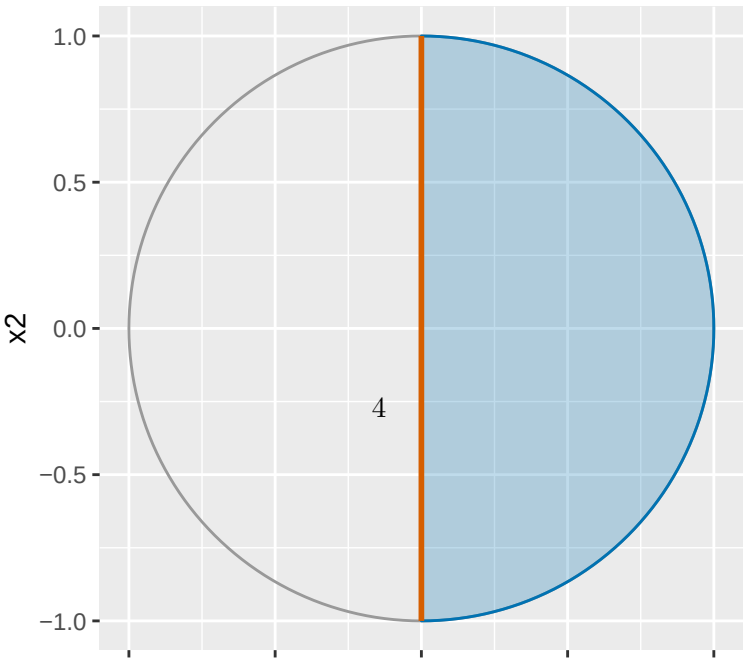
Python


```python
import matplotlib.pyplot as plt
import numpy as np

cs_angle_fc = np.linspace(0, 2 * np.pi, 201)
cs_x1_fc = np.array([-0.7, 0.6])
cs_x2_fc = np.array([-0.4, 0.5])
```

R

```
ci_angle_ci <- seq(-pi / 2, pi / 2, length.out = 101)
ci_half_x1_ci <- cos(ci_angle_ci)
ci_half_x2_ci <- sin(ci_angle_ci)
ci_half_ci <- tibble::tibble(x1 = ci_half_x1_ci, x2 = ci_half_x2_ci)
ci_full_angle_ci <- seq(0, 2 * pi, length.out = 201)
ci_full_x1_ci <- cos(ci_full_angle_ci)
ci_full_x2_ci <- sin(ci_full_angle_ci)
ci_full_ci <- tibble::tibble(x1 = ci_full_x1_ci, x2 = ci_full_x2_ci)
ci_edge_ci <- tibble::tibble(x1 = 0, x2 = -1, x1_end = 0, x2_end = 1)
ci_aes_ci <- ggplot2::aes(x1, x2)
ci_seg_aes_ci <- ggplot2::aes(xend = x1_end, yend = x2_end)
ci_full_ci |>
  ggplot2::ggplot() +
  ci_aes_ci +
  ggplot2::geom_path(colour = "grey60") +
  ggplot2::geom_polygon(
    data = ci_half_ci, fill = "#0072B2", alpha = 0.25, colour = "#0072B2"
  ) +
  ggplot2::geom_segment(
    data = ci_edge_ci, mapping = ci_seg_aes_ci,
    colour = "#D55E00", linewidth = 1
  ) +
  ggplot2::coord_fixed() +
  ggplot2::labs(x = "x1", y = "x2")
```



Definition 0.3 (Strictly convex function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}$ is **strictly convex** if $f(t\underset{\sim}{x} + (1-t)\underset{\sim}{y}) < tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y})$ for all $\underset{\sim}{x} \neq \underset{\sim}{y}$ in $\mathbb{R}^p$ and all $t \in (0,1)$.

Definition 0.4 (Concave function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}$ is **concave** if $-f$ is convex.

Definition 0.5 (Strictly concave function)**.** A function $f : \mathbb{R}^p \to \mathbb{R}$ is **strictly concave** if $-f$ is strictly convex (Definition 0.3).

Example 0.5 (Strictly convex, and convex but not strictly)**.**

- $f(x) = x^2$ is strictly convex:

$$\begin{aligned} tx^2 + (1-t)y^2 - (tx + (1-t)y)^2 &= tx^2 + (1-t)y^2 - t^2x^2 - 2t(1-t)xy - (1-t)^2y^2 && \text{(expand the squa} \\ &= t(1-t)x^2 - 2t(1-t)xy + t(1-t)y^2 && (t - t^2 = t(1-t) \\ &= t(1-t)(x-y)^2, && \text{(factor)} \end{aligned}$$

  which is positive when $x \neq y$ and $t \in (0,1)$. At $t \in \{0,1\}$ or $x = y$ it is 0, so x^2 is also convex.

- $f(x) = 2x + 1$ is convex but not strictly convex:

$$\begin{aligned} f(tx + (1-t)y) &= 2tx + 2(1-t)y + 1 \\ &= tf(x) + (1-t)f(y) \end{aligned}$$

  for every x, y, t, so the inequality holds, but never strictly. It is also concave.

Theorem 0.2 (An open ball is convex)**.** *Let $\underset{\sim}{x}_0 \in \mathbb{R}^p$ and $r > 0$. The open ball of radius r around $\underset{\sim}{x}_0$ is a convex set.*

Proof

Proof. Call the open ball B, and let $\underset{\sim}{x}, \underset{\sim}{y} \in B$ and $t \in [0,1]$. The triangle inequality and the absolute homogeneity of the norm give

$$\begin{aligned} \|t\underset{\sim}{x} + (1-t)\underset{\sim}{y} - \underset{\sim}{x}_0\| &= \|t(\underset{\sim}{x} - \underset{\sim}{x}_0) + (1-t)(\underset{\sim}{y} - \underset{\sim}{x}_0)\| && (\underset{\sim}{x}_0 = t\underset{\sim}{x}_0 + (1-t)\underset{\sim}{x}_0) \\ &\leq t\|\underset{\sim}{x} - \underset{\sim}{x}_0\| + (1-t)\|\underset{\sim}{y} - \underset{\sim}{x}_0\| && \text{(triangle inequality and } t, 1-t \geq 0) \\ &< r, && \text{(weighted mean of two numbers less than } r) \end{aligned}$$

so $t\underset{\sim}{x} + (1-t)\underset{\sim}{y} \in B$. □

Definition 0.6 (Convex function on a set)**.** Let $K \subseteq \mathbb{R}^p$ be a convex set. A function $f : \mathbb{R}^p \to \mathbb{R}$ is **convex on** K if

$$f(t\underset{\sim}{x} + (1-t)\underset{\sim}{y}) \leq tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y})$$

for all $\underset{\sim}{x}, \underset{\sim}{y} \in K$ and all $t \in [0,1]$.

Example 0.6 (A function convex on an interval but not on the whole line)**.** Let $f(x) = \min(x^2, 1)$.

- f is convex on $[-1,1]$, which is the closed ball of radius 1 around 0 and so convex by Example 0.1. For $x, y \in [-1,1]$ and $t \in [0,1]$, the point $tx + (1-t)y$ is also in $[-1,1]$, where f agrees with x^2, so the defining inequality of f is the defining inequality of x^2, which holds by Example 0.5.
- f is not convex on all of $\mathbb{R}$: with $x = 0$, $y = 2$ and $t = \frac{1}{2}$,

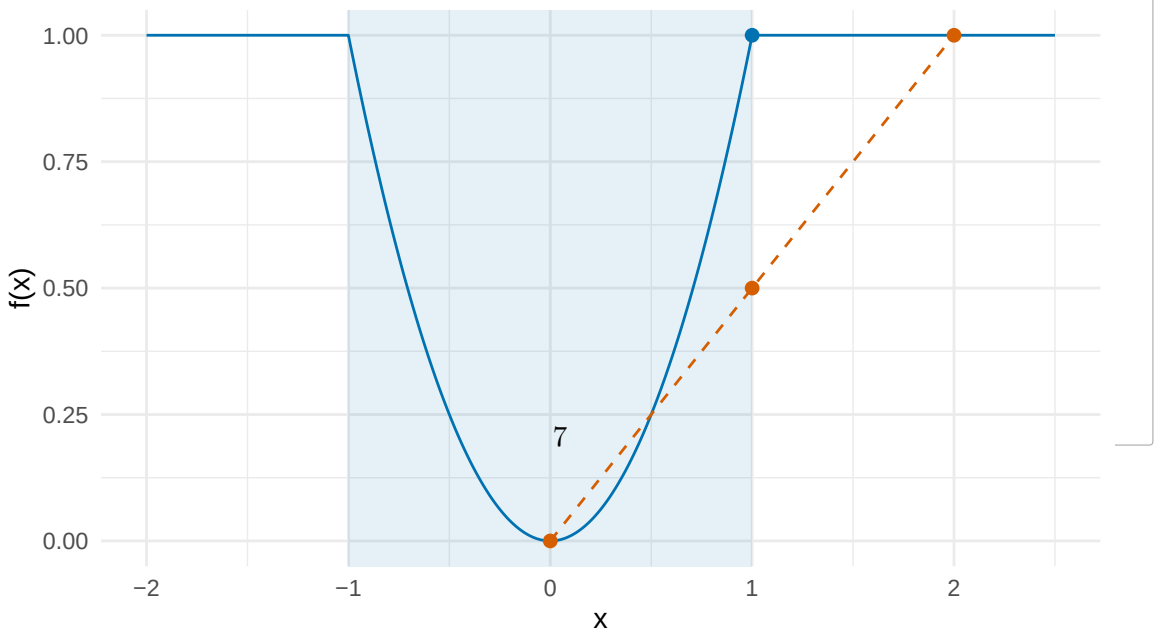
$$\begin{aligned} f\left(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot 2\right) &= f(1) && \left(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot 2 = 1\right) \\ &= 1, && (\min(1,1) = 1) \end{aligned}$$

  which is greater than $\frac{1}{2}f(0) + \frac{1}{2}f(2) = \frac{1}{2}$, since $f(0) = 0$ and $f(2) = 1$.

In Figure 3, the chord from 0 to 2 passes below the graph at 1, while every chord with both ends in $[-1,1]$ lies on or above it.

R

```r
co_x_co <- seq(-2, 2.5, length.out = 451)
co_square_co <- co_x_co^2
co_f_co <- pmin(co_square_co, 1)
co_graph_co <- tibble::tibble(x = co_x_co, f = co_f_co)
co_chord_x_co <- c(0, 2)
co_chord_f_co <- c(0, 1)
co_chord_co <- tibble::tibble(x = co_chord_x_co, f = co_chord_f_co)
co_mid_x_co <- c(1, 1)
co_mid_f_co <- c(0.5, 1)
co_mid_co <- tibble::tibble(x = co_mid_x_co, f = co_mid_f_co)
co_mid_cols_co <- c("#D55E00", "#0072B2")
co_graph_co |>
  ggplot2::ggplot() +
  ggplot2::aes(x = x, y = f) +
  ggplot2::annotate(
    "rect",
    xmin = -1, xmax = 1, ymin = -Inf, ymax = Inf,
    fill = "#0072B2", alpha = 0.1
  ) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = co_chord_co, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = co_chord_co, colour = "#D55E00", size = 2) +
  ggplot2::geom_point(data = co_mid_co, colour = co_mid_cols_co, size = 2) +
  ggplot2::labs(x = "x", y = "f(x)") +
  ggplot2::theme_minimal()
```



Definition 0.7 (Function convex at a point)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ and let $\underset{\sim}{x}_0 \in \mathbb{R}^p$. f is **convex at** $\underset{\sim}{x}_0$ if there is an $r > 0$ such that f is convex on the open ball of radius r around $\underset{\sim}{x}_0$.

Example 0.7 (A function convex at one point and not at another)**.** Let $f(x) = \min(x^2, 1)$, as in Example 0.6.

- f is convex at 0: the open ball of radius 1 around 0 is $(-1, 1)$, a subset of $[-1, 1]$, and f is convex on $[-1, 1]$ by Example 0.6, so the defining inequality holds for all $x, y \in (-1, 1)$ as well.
- f is not convex at 1. Take any $r > 0$, let $\varepsilon = \frac{1}{2}\min(r, 1)$, and take $x = 1 - \varepsilon$, $y = 1 + \varepsilon$ and $t = \frac{1}{2}$, so that x and y are in the open ball of radius r around 1 and $tx + (1 - t)y = 1$. Then $f(1) = 1$, while

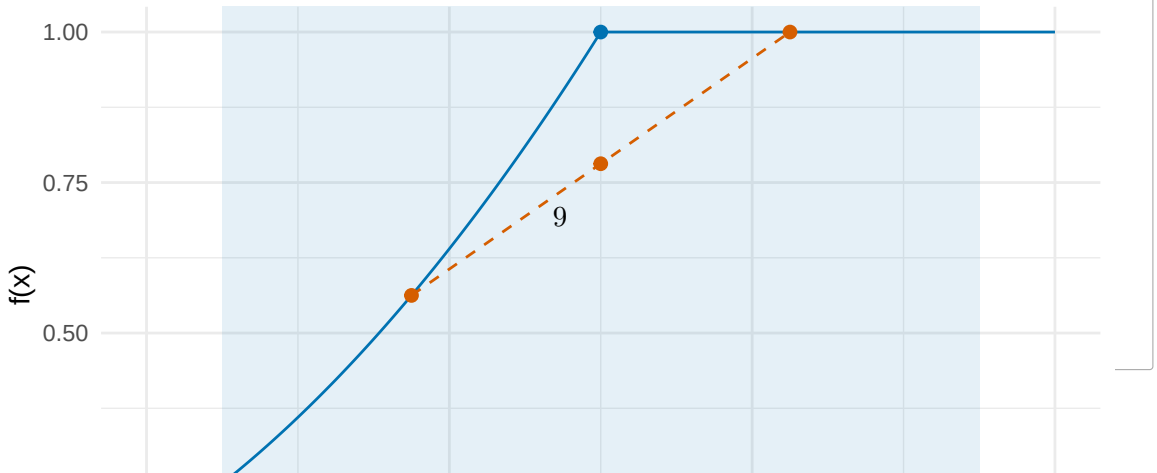
$$\begin{aligned} \tfrac{1}{2}f(1-\varepsilon) + \tfrac{1}{2}f(1+\varepsilon) &= \tfrac{1}{2}(1-\varepsilon)^2 + \tfrac{1}{2}\cdot 1 && (0 < 1-\varepsilon < 1 < 1+\varepsilon) \\ &< \tfrac{1}{2} + \tfrac{1}{2} && ((1-\varepsilon)^2 < 1) \\ &= 1, && (\text{add}) \end{aligned}$$

  so the defining inequality fails on every open ball around 1.

In Figure 4, the chord from $1 - \varepsilon$ to $1 + \varepsilon$ passes below the graph at 1.

R

```
ca_x_ca <- seq(0.4, 1.6, length.out = 241)
ca_square_ca <- ca_x_ca^2
ca_f_ca <- pmin(ca_square_ca, 1)
ca_graph_ca <- tibble::tibble(x = ca_x_ca, f = ca_f_ca)
ca_ends_ca <- c(0.75, 1.25)
ca_end_squares_ca <- ca_ends_ca^2
ca_end_f_ca <- pmin(ca_end_squares_ca, 1)
ca_chord_ca <- tibble::tibble(x = ca_ends_ca, f = ca_end_f_ca)
ca_chord_mid_ca <- mean(ca_end_f_ca)
ca_mid_x_ca <- c(1, 1)
ca_mid_f_ca <- c(ca_chord_mid_ca, 1)
ca_mid_ca <- tibble::tibble(x = ca_mid_x_ca, f = ca_mid_f_ca)
ca_mid_cols_ca <- c("#D55E00", "#0072B2")
ca_graph_ca |>
  ggplot2::ggplot() +
  ggplot2::aes(x = x, y = f) +
  ggplot2::annotate(
    "rect",
    xmin = 0.5, xmax = 1.5, ymin = -Inf, ymax = Inf,
    fill = "#0072B2", alpha = 0.1
  ) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = ca_chord_ca, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = ca_chord_ca, colour = "#D55E00", size = 2) +
  ggplot2::geom_point(data = ca_mid_ca, colour = ca_mid_cols_ca, size = 2) +
  ggplot2::labs(x = "x", y = "f(x)") +
  ggplot2::theme_minimal()
```



Theorem 0.3 (The maximum of convex functions is convex)**.** *If f_1 and f_2 are convex functions on $\mathbb{R}^p$, then $f(\underset{\sim}{x}) \stackrel{def}{=} \max\{f_1(\underset{\sim}{x}), f_2(\underset{\sim}{x})\}$ is convex.*

Proof

Proof. Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ and $t \in [0,1]$. For $i = 1, 2$,

$$\begin{aligned} f_i(t\underset{\sim}{x} + (1-t)\underset{\sim}{y}) &\leq tf_i(\underset{\sim}{x}) + (1-t)f_i(\underset{\sim}{y}) \quad (f_i \text{ is convex}) \\ &\leq tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y}). \quad (f_i \leq f, \text{ and } t, 1-t \geq 0) \end{aligned}$$

Both f_1 and f_2 at $t\underset{\sim}{x} + (1-t)\underset{\sim}{y}$ are at most $tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y})$, so their maximum $f(t\underset{\sim}{x} + (1-t)\underset{\sim}{y})$ is too. □

Example 0.8 (A maximum is convex; a minimum need not be)**.**

- $f_1(x) = x$ is convex, by the computation for $2x + 1$ in Example 0.5 with 2 and 1 replaced by 1 and 0, and $f_2(x) = x^2$ is convex by the same example, so $\max\{x, x^2\}$ is convex by Theorem 0.3.
- The minimum of two convex functions need not be convex: $g(x) = \min\{x^2, (x-2)^2\}$ has

  $$\begin{aligned} g(0) &= g(2) \\ &= 0, \end{aligned}$$

  but at the midpoint

  $$\begin{aligned} g(1) &= \min\{1, 1\} \\ &= 1 \\ &> \tfrac{1}{2}g(0) + \tfrac{1}{2}g(2) \\ &= 0. \end{aligned}$$

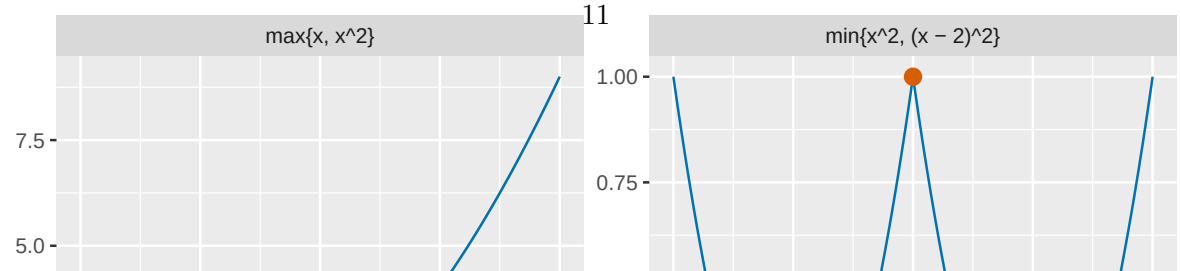
Definition 0.8 (Epigraph)**.** The **epigraph** [3] of $f : \mathbb{R}^p \to \mathbb{R}$ is the set of points on or over its graph,

$$\operatorname{epi} f \stackrel{\text{def}}{=} \{(\underset{\sim}{x}, s) : \underset{\sim}{x} \in \mathbb{R}^p,\ s \in \mathbb{R},\ f(\underset{\sim}{x}) \leq s\}.$$

A pair $(\underset{\sim}{x}, s)$ is treated as the $(p+1)$-vector with entries $x_1, \ldots, x_p, s$, so $\operatorname{epi} f \subseteq \mathbb{R}^{p+1}$, and sums and multiples of pairs are taken entry by entry.

R

```
cm_x_fm <- seq(-1, 3, length.out = 201)
cm_levels_fm <- c("max{x, x^2}", "min{x^2, (x - 2)^2}")
cm_max_value_fm <- pmax(cm_x_fm, cm_x_fm^2)
cm_min_value_fm <- pmin(cm_x_fm^2, (cm_x_fm - 2)^2)
cm_max_fm <- tibble::tibble(
  x = cm_x_fm, value = cm_max_value_fm, panel = cm_levels_fm[1]
)
cm_min_fm <- tibble::tibble(
  x = cm_x_fm, value = cm_min_value_fm, panel = cm_levels_fm[2]
)
cm_main_fm <- dplyr::bind_rows(cm_max_fm, cm_min_fm)
cm_main_fm$panel <- factor(cm_main_fm$panel, levels = cm_levels_fm)
cm_square_value_fm <- cm_x_fm^2
cm_line_fm <- tibble::tibble(
  x = cm_x_fm, value = cm_x_fm, panel = cm_levels_fm[1], part = "x"
)
cm_square_fm <- tibble::tibble(
  x = cm_x_fm, value = cm_square_value_fm, panel = cm_levels_fm[1],
  part = "x^2"
)
cm_parts_fm <- dplyr::bind_rows(cm_line_fm, cm_square_fm)
cm_parts_fm$panel <- factor(cm_parts_fm$panel, levels = cm_levels_fm)
cm_chord_fm <- tibble::tibble(
  x = 0, xend = 2, value = 0, panel = cm_levels_fm[2]
)
cm_chord_fm$panel <- factor(cm_chord_fm$panel, levels = cm_levels_fm)
cm_mid_fm <- tibble::tibble(x = 1, value = 1, panel = cm_levels_fm[2])
cm_mid_fm$panel <- factor(cm_mid_fm$panel, levels = cm_levels_fm)
cm_aes_fm <- ggplot2::aes(x, value)
cm_parts_aes_fm <- ggplot2::aes(group = part)
cm_chord_aes_fm <- ggplot2::aes(xend = xend, yend = value)
ggplot2::ggplot(cm_main_fm, cm_aes_fm) +
  ggplot2::geom_line(
    data = cm_parts_fm, mapping = cm_parts_aes_fm,
    colour = "grey60", linewidth = 0.4
  ) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_segment(
    data = cm_chord_fm, mapping = cm_chord_aes_fm, colour = "#D55E00"
  ) +
  ggplot2::geom_point(data = cm_mid_fm, colour = "#D55E00", size = 3) +
  ggplot2::facet_wrap(~panel, scales = "free_y") +
  ggplot2::labs(x = "x", y = NULL)
```



Example 0.9 (The epigraph of x^2)**.** For $f(x) = x^2$, $\operatorname{epi} f = \{(x, s) : x^2 \le s\}$, the points (x, s) on the parabola $s = x^2$ or on the side of it where s is larger. $(1, 2)$ is in it, since $1^2 \le 2$; $(2, 1)$ is not, since $2^2 > 1$.

Theorem 0.4 (Convexity through the epigraph)**.** *$f : \mathbb{R}^p \to \mathbb{R}$ is convex if and only if its epigraph (Definition 0.8) is convex (Definition 0.1).*

Proof

Proof. **If.** Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$ and $t \in [0, 1]$. The points $(\underset{\sim}{x}, f(\underset{\sim}{x}))$ and $(\underset{\sim}{y}, f(\underset{\sim}{y}))$ are in $\operatorname{epi} f$, since $f(\underset{\sim}{x}) \le f(\underset{\sim}{x})$ and $f(\underset{\sim}{y}) \le f(\underset{\sim}{y})$. Since $\operatorname{epi} f$ is convex, it contains

$$t\,(\underset{\sim}{x}, f(\underset{\sim}{x})) + (1-t)\,(\underset{\sim}{y}, f(\underset{\sim}{y})) = (t\underset{\sim}{x} + (1-t)\underset{\sim}{y},\ tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y})),$$

and by Definition 0.8 that membership means $f(t\underset{\sim}{x} + (1-t)\underset{\sim}{y}) \le tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y})$.
Only if. Let $(\underset{\sim}{x}, s)$ and $(\underset{\sim}{y}, u)$ be in $\operatorname{epi} f$, so $f(\underset{\sim}{x}) \le s$ and $f(\underset{\sim}{y}) \le u$, and let $t \in [0, 1]$. Then

$$\begin{aligned} f(t\underset{\sim}{x} + (1-t)\underset{\sim}{y}) &\le tf(\underset{\sim}{x}) + (1-t)f(\underset{\sim}{y}) && (f \text{ is convex}) \\ &\le ts + (1-t)u, && (f(\underset{\sim}{x}) \le s,\ f(\underset{\sim}{y}) \le u, \text{ and } t, 1-t \ge 0) \end{aligned}$$

so $t\,(\underset{\sim}{x}, s) + (1-t)\,(\underset{\sim}{y}, u) = (t\underset{\sim}{x} + (1-t)\underset{\sim}{y},\ ts + (1-t)u)$ is in $\operatorname{epi} f$. □

Example 0.10 (Two epigraphs)**.**

- The epigraph of $f(x) = |x|$ is $\{(x, s) : |x| \le s\}$, a wedge. It is the intersection of the half-planes $\{(x, s) : s - x \ge 0\}$ and $\{(x, s) : s + x \ge 0\}$. Each half-plane is convex: if $s_1 - x_1 \ge 0$ and $s_2 - x_2 \ge 0$, then $(ts_1 + (1-t)s_2) - (tx_1 + (1-t)x_2) = t(s_1 - x_1) + (1-t)(s_2 - x_2) \ge 0$, and the same with $+$ for the other. So the wedge is convex (Theorem 0.1), and $|x|$ is convex by Theorem 0.4.
- The epigraph of $g(x) = -x^2$ is not convex: $(-1, -1)$ and $(1, -1)$ are in it, since $g(\pm 1) = -1$, but their midpoint $(0, -1)$ is not, since $g(0) = 0 > -1$. So $-x^2$ is not convex.

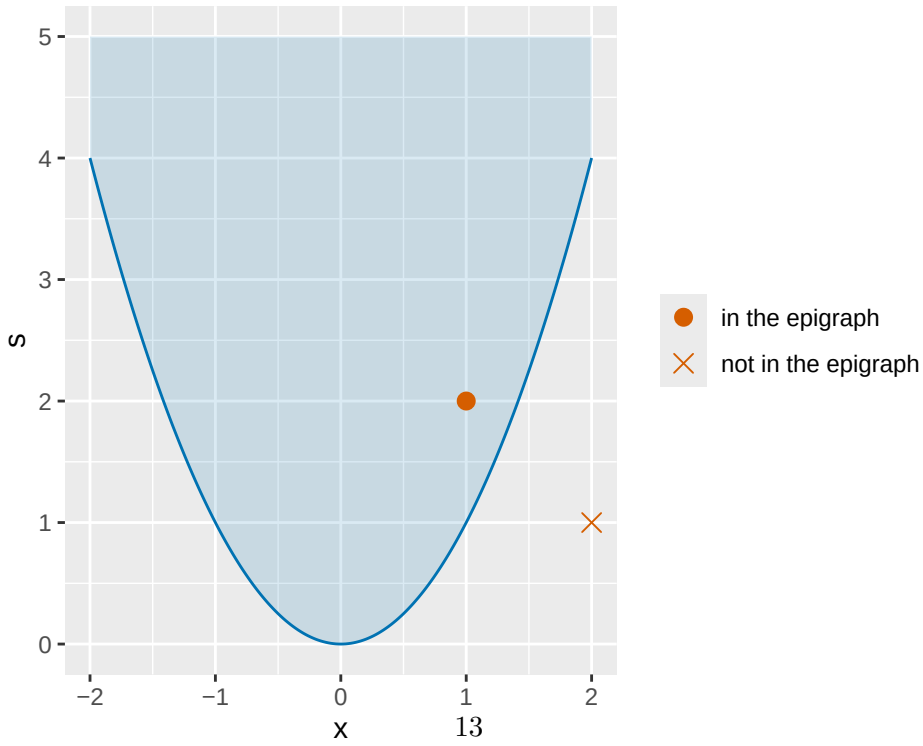
Definition 0.9 (Tangent plane)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient at $\underset{\sim}{y}$. The **tangent plane** [4] to the graph of f at $\underset{\sim}{y}$ is the graph of the function

$$\underset{\sim}{x} \mapsto f(\underset{\sim}{y}) + \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^{\top} (\underset{\sim}{x} - \underset{\sim}{y}),$$

which takes the value $f(\underset{\sim}{y})$ at $\underset{\sim}{x} = \underset{\sim}{y}$ and has the same gradient as f there.

R

```r
ep_x_fe <- seq(-2, 2, length.out = 201)
ep_curve_fe <- tibble::tibble(x = ep_x_fe, s = ep_x_fe^2, top = 5)
ep_marks_x_fe <- c(1, 2)
ep_marks_s_fe <- c(2, 1)
ep_marks_kind_fe <- c("in the epigraph", "not in the epigraph")
ep_marks_fe <- tibble::tibble(
  x = ep_marks_x_fe, s = ep_marks_s_fe, kind = ep_marks_kind_fe
)
ep_aes_fe <- ggplot2::aes(x, s)
ep_ribbon_aes_fe <- ggplot2::aes(ymin = s, ymax = top)
ep_marks_aes_fe <- ggplot2::aes(shape = kind)
ep_shapes_fe <- c("in the epigraph" = 16, "not in the epigraph" = 4)
ggplot2::ggplot(ep_curve_fe, ep_aes_fe) +
  ggplot2::geom_ribbon(
    mapping = ep_ribbon_aes_fe, fill = "#0072B2", alpha = 0.15
  ) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(
    data = ep_marks_fe, mapping = ep_marks_aes_fe,
    colour = "#D55E00", size = 3
  ) +
  ggplot2::scale_shape_manual(values = ep_shapes_fe) +
  ggplot2::labs(x = "x", y = "s", shape = NULL)
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np
```

Example 0.11 (A tangent line and a tangent plane)**.**

- For $f(x) = x^2$ at $y = 1$, $f(1) = 1$ and $f'(1) = 2$, so the tangent line is

$$\begin{aligned} s &= 1 + 2\,(x - 1) && \text{(substitute } f(1) \text{ and } f'(1)\text{)} \\ &= 1 + 2x - 2 && \text{(distribute)} \\ &= 2x - 1. && (1 - 2 = -1) \end{aligned}$$

- For $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ at $\underset{\sim}{y} = (1, 1)^\top$, $f(\underset{\sim}{y}) = 2$ and the gradient is $\nabla f(\underset{\sim}{y}) = (2y_1,\ 2y_2)^\top = (2, 2)^\top$, so the tangent plane is

$$\begin{aligned} s &= 2 + 2\,(x_1 - 1) + 2\,(x_2 - 1) && \text{(substitute } f(\underset{\sim}{y}) \text{ and the gradient)} \\ &= 2 + 2x_1 - 2 + 2x_2 - 2 && \text{(distribute)} \\ &= 2x_1 + 2x_2 - 2. && (2 - 2 - 2 = -2) \end{aligned}$$

Figure 7 shows the tangent plane of the second bullet.

R

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Python

```
import matplotlib.pyplot as plt
import numpy as np

tp_x1_py, tp_x2_py = np.meshgrid(np.linspace(-1, 3, 41),
                                 np.linspace(-1, 3, 41))
tp_bowl_py = tp_x1_py**2 + tp_x2_py**2
tp_plane_py = 2 * tp_x1_py + 2 * tp_x2_py - 2
fig = plt.figure(figsize=(6, 4.5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(tp_x1_py, tp_x2_py, tp_bowl_py, color="#0072B2", alpha=0.6)
ax.plot_surface(tp_x1_py, tp_x2_py, tp_plane_py, color="#D55E00", alpha=0.4)
ax.scatter([1], [1], [2], color="black", label="point of tangency")
ax.set_xlabel("x1")
ax.set_ylabel("x2")
ax.set_zlabel("f(x)")
ax.legend(loc="upper right")
plt.show();
```

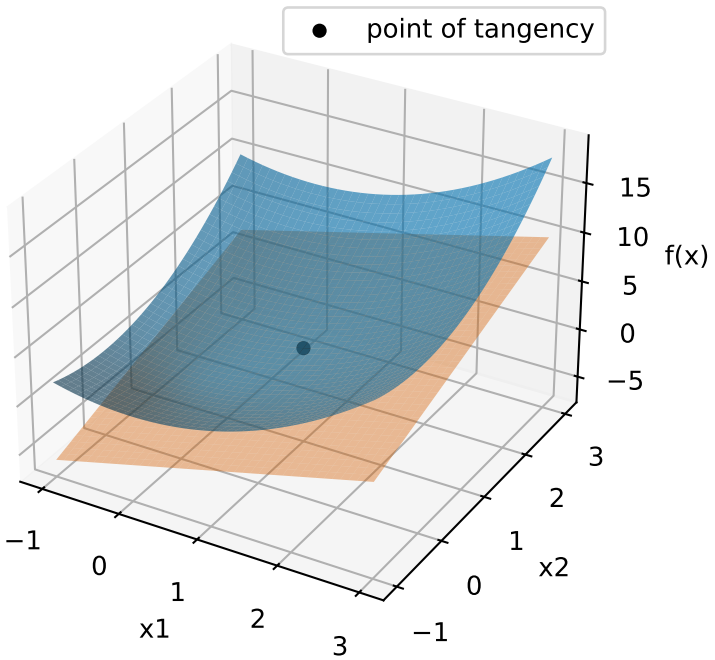


Figure 7: The tangent plane (vermilion) of the bowl $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ of Example 0.11 touches it at $(1, 1)^\top$ and lies below it.

Remark (The tangent plane in one variable)

Remark 0.1 (The tangent plane in one variable). For $p = 1$ the tangent plane is the tangent line $s = f(y) + f'(y)\,(x - y)$, the line through $(y, f(y))$ with slope $f'(y)$.

Theorem 0.5 (First-order characterization of convexity)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$. Then f is convex if and only if*

$$f(\underset{\sim}{x}) \geq f(\underset{\sim}{y}) + \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^{\top} (\underset{\sim}{x} - \underset{\sim}{y}) \qquad \text{for all } \underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p : \tag{1}$$

every tangent plane (Definition 0.9) lies on or under the graph. If the inequality is strict whenever $\underset{\sim}{x} \neq \underset{\sim}{y}$, then f is strictly convex (Definition 0.3).

Proof

Proof. **Only if.** Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$, let $\underset{\sim}{d} \overset{\text{def}}{=} \underset{\sim}{x} - \underset{\sim}{y}$, and let $g(t) \overset{\text{def}}{=} f(\underset{\sim}{y} + t\underset{\sim}{d})$. The inner function $t \mapsto \underset{\sim}{y} + t\underset{\sim}{d}$ has derivative $\underset{\sim}{d}^{\top}$, so the vector chain rule gives

$$\begin{aligned} g'(0) &= \underset{\sim}{d}^{\top} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y}) && \text{(vector chain rule)} \\ &= \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^{\top} \underset{\sim}{d}. && \text{(a } 1 \times 1 \text{ matrix equals its transpose)} \end{aligned}$$

For $t \in (0, 1]$, $\underset{\sim}{y} + t\underset{\sim}{d} = t\underset{\sim}{x} + (1 - t)\underset{\sim}{y}$, so

$$\begin{aligned} g(t) - g(0) &= f(t\underset{\sim}{x} + (1 - t)\underset{\sim}{y}) - f(\underset{\sim}{y}) && \text{(definition of } g) \\ &\leq t f(\underset{\sim}{x}) + (1 - t) f(\underset{\sim}{y}) - f(\underset{\sim}{y}) && (f \text{ is convex}) \\ &= t\,(f(\underset{\sim}{x}) - f(\underset{\sim}{y})), && \text{(collect terms)} \end{aligned}$$

and dividing by $t > 0$, $\dfrac{g(t) - g(0)}{t} \leq f(\underset{\sim}{x}) - f(\underset{\sim}{y})$. As $t \to 0$ from the right, the left side tends to $g'(0)$ (g is differentiable at 0, so the one-sided limit equals the two-sided derivative), and a limit of numbers that are all at most $f(\underset{\sim}{x}) - f(\underset{\sim}{y})$ is at most $f(\underset{\sim}{x}) - f(\underset{\sim}{y})$. So $\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^{\top} (\underset{\sim}{x} - \underset{\sim}{y}) \leq f(\underset{\sim}{x}) - f(\underset{\sim}{y})$, which rearranges to Equation 1.

If. Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$, $t \in [0, 1]$, $\underset{\sim}{z} \overset{\text{def}}{=} t\underset{\sim}{x} + (1 - t)\underset{\sim}{y}$, and $\underset{\sim}{g} \overset{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{z})$. Applying Equation 1 at the point $\underset{\sim}{z}$, once toward $\underset{\sim}{x}$ and once toward $\underset{\sim}{y}$,

$$f(\underset{\sim}{x}) \geq f(\underset{\sim}{z}) + \underset{\sim}{g}^{\top} (\underset{\sim}{x} - \underset{\sim}{z}), \qquad f(\underset{\sim}{y}) \geq f(\underset{\sim}{z}) + \underset{\sim}{g}^{\top} (\underset{\sim}{y} - \underset{\sim}{z}).$$

Multiplying the first by $t \geq 0$ and the second by $1 - t \geq 0$ and adding,

$$
\begin{aligned}
t f(\underset{\sim}{x}) + (1-t) f(\underset{\sim}{y}) &\geq t(f(\underset{\sim}{z}) + \underset{\sim}{g}^\top(\underset{\sim}{x} - \underset{\sim}{z})) + (1-t)(f(\underset{\sim}{z}) + \underset{\sim}{g}^\top(\underset{\sim}{y} - \underset{\sim}{z})) && \text{(multiply by } t, 1-t \geq 0 \text{ and add)} \\
&= (t + (1-t))\, f(\underset{\sim}{z}) + t\, \underset{\sim}{g}^\top(\underset{\sim}{x} - \underset{\sim}{z}) + (1-t)\, \underset{\sim}{g}^\top(\underset{\sim}{y} - \underset{\sim}{z}) && \text{(distribute } t \text{ and } 1-t) \\
&= f(\underset{\sim}{z}) + t\, \underset{\sim}{g}^\top(\underset{\sim}{x} - \underset{\sim}{z}) + (1-t)\, \underset{\sim}{g}^\top(\underset{\sim}{y} - \underset{\sim}{z}) && (t + (1-t) = 1) \\
&= f(\underset{\sim}{z}) + \underset{\sim}{g}^\top(t(\underset{\sim}{x} - \underset{\sim}{z}) + (1-t)(\underset{\sim}{y} - \underset{\sim}{z})) && \text{(factor out } \underset{\sim}{g}^\top\text{; matrix products dis} \\
&= f(\underset{\sim}{z}) + \underset{\sim}{g}^\top(t\underset{\sim}{x} + (1-t)\underset{\sim}{y} - \underset{\sim}{z}) && \text{(collect terms)} \\
&= f(\underset{\sim}{z}). && \text{(definition of } \underset{\sim}{z})
\end{aligned}
$$

That inequality is the defining inequality of a convex function. If Equation 1 is strict for distinct points, and $\underset{\sim}{x} \neq \underset{\sim}{y}$ and $t \in (0,1)$, then:

- $\underset{\sim}{z}$ differs from both $\underset{\sim}{x}$ and $\underset{\sim}{y}$, since $\underset{\sim}{x} - \underset{\sim}{z} = (1-t)(\underset{\sim}{x} - \underset{\sim}{y}) \neq \underset{\sim}{0}$ and $\underset{\sim}{y} - \underset{\sim}{z} = t(\underset{\sim}{y} - \underset{\sim}{x}) \neq \underset{\sim}{0}$;
- so both inequalities at $\underset{\sim}{z}$ are strict;
- both multipliers t and $1-t$ are positive, so the first step of the display is strict, and f is strictly convex.

□

Example 0.12 (Tangent lines under a parabola)**.** For $f(x) = x^2$, the right side of Equation 1 is $y^2 + 2y(x-y) = 2xy - y^2$, and

$$
\begin{aligned}
f(x) - (2xy - y^2) &= x^2 - 2xy + y^2 \\
&= (x-y)^2 \\
&\geq 0,
\end{aligned}
$$

with equality only at $x = y$, confirming that x^2 is strictly convex. For $g(x) = -x^2$ the inequality fails: at $y = 0$ the tangent line is $s = 0$, and $g(1) = -1 < 0$.

Theorem 0.6 (Second-derivative test for convexity)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$.*

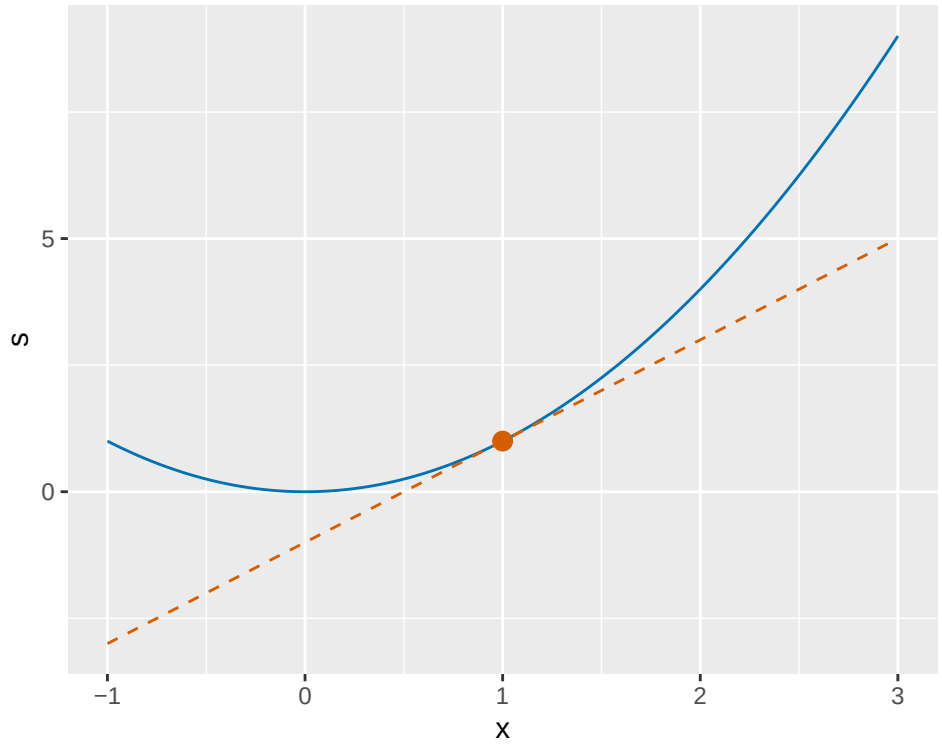
1. *If $\mathbf{H}_f(\underset{\sim}{x})$ is positive semidefinite for every $\underset{\sim}{x}$, then f is convex.*
2. *If $\mathbf{H}_f(\underset{\sim}{x})$ is positive definite for every $\underset{\sim}{x}$, then f is strictly convex (Definition 0.3).*

Proof

Proof. Let $\underset{\sim}{x}, \underset{\sim}{y} \in \mathbb{R}^p$. By Theorem 6 in Critical Points and Optimality Conditions with $\underset{\sim}{z} = \underset{\sim}{y}$ and $\underset{\sim}{h} = \underset{\sim}{x} - \underset{\sim}{y}$, there is a $\tau \in (0,1)$ such that, with $\underset{\sim}{w} \overset{\text{def}}{=} \underset{\sim}{y} + \tau(\underset{\sim}{x} - \underset{\sim}{y})$,

R

```r
tl_x_ft <- seq(-1, 3, length.out = 201)
tl_curve_ft <- tibble::tibble(x = tl_x_ft, s = tl_x_ft^2)
tl_line_ft <- tibble::tibble(x = tl_x_ft, s = 2 * tl_x_ft - 1)
tl_point_ft <- tibble::tibble(x = 1, s = 1)
tl_aes_ft <- ggplot2::aes(x, s)
ggplot2::ggplot(tl_curve_ft, tl_aes_ft) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_line(
    data = tl_line_ft, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::geom_point(data = tl_point_ft, colour = "#D55E00", size = 3) +
  ggplot2::labs(x = "x", y = "s")
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np

tl_x_ft = np.linspace(-1, 3, 201)
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(tl_x_ft, tl_x_ft**2, color="#0072B2", label="f(x) = x^2")
ax.plot(tl_x_ft, 2 * tl_x_ft - 1, "--", color="#D55E00",
        label="tangent line at 1")
ax.plot(1, 1, "o", color="#D55E00", markersize=8)
ax.set_xlabel("x")
ax.set_ylabel("s")
ax.legend(loc="upper center")
plt.show()
```

$$f(\underset{\sim}{x}) = f(\underset{\sim}{y}) + \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^\top (\underset{\sim}{x} - \underset{\sim}{y}) + \frac{1}{2} (\underset{\sim}{x} - \underset{\sim}{y})^\top \mathbf{H}_f(\underset{\sim}{w}) (\underset{\sim}{x} - \underset{\sim}{y}) \quad \text{(Theorem 6 in Critical Points and Optimality}$$
$$\geq f(\underset{\sim}{y}) + \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{y})\right)^\top (\underset{\sim}{x} - \underset{\sim}{y}). \quad (\mathbf{H}_f(\underset{\sim}{w}) \text{ is positive semidefinite})$$

This inequality is Equation 1, so f is convex by Theorem 0.5. Under the hypothesis of part 2 and with $\underset{\sim}{x} \neq \underset{\sim}{y}$, the quadratic-form term is positive, so the inequality is strict, and f is strictly convex by the last sentence of Theorem 0.5. □

Example 0.13 (Using the Hessian to show convexity)**.**

- For a symmetric positive semidefinite $\mathbf{A}$, $f(\underset{\sim}{x}) = \underset{\sim}{x}^\top \mathbf{A} \underset{\sim}{x}$ has Hessian $2\mathbf{A}$ (by the Hessian of a quadratic form), which is positive semidefinite because $\underset{\sim}{h}^\top (2\mathbf{A}) \underset{\sim}{h} = 2\, \underset{\sim}{h}^\top \mathbf{A} \underset{\sim}{h} \geq 0$; so f is convex.
- $f(\underset{\sim}{x}) = x_1^2 + x_2^2 - 2x_1$ has Hessian $2\mathbf{I}_2$ (Example 16 in Critical Points and Optimality Conditions), which is positive definite, so f is strictly convex.
- $f(\underset{\sim}{x}) = e^{2x_1 + x_2} - x_1$ has Hessian $e^u \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ with $u = 2x_1 + x_2$ (the Hessian example), and

$$\begin{aligned} \underset{\sim}{h}^\top \mathbf{H}_f(\underset{\sim}{x})\, \underset{\sim}{h} &= e^u (4h_1^2 + 4h_1 h_2 + h_2^2) \\ &= e^u (2h_1 + h_2)^2 \\ &\geq 0, \end{aligned}$$

  so f is convex. The Hessian is not positive definite ($\underset{\sim}{h} = (1, -2)^\top$ gives 0), so part 2 does not apply, and in fact f is not strictly convex: along $\underset{\sim}{x} = t\,(1, -2)^\top$,

$$\begin{aligned} u &= 2t - 2t \\ &= 0 \end{aligned}$$

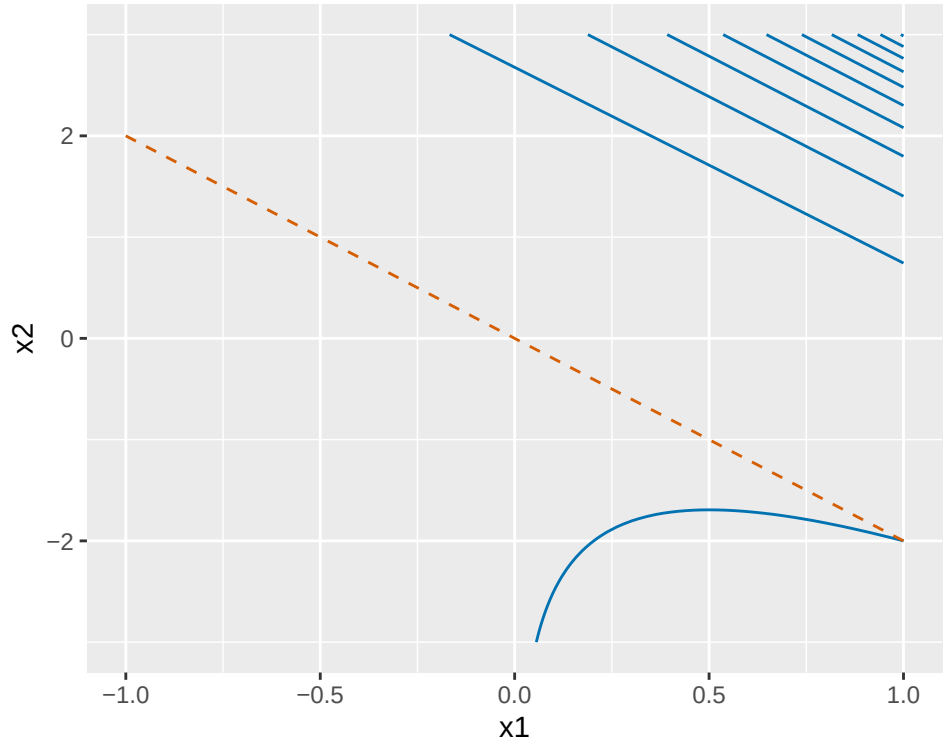
  and $f = 1 - t$ is linear in t, so the convexity inequality holds with equality there.

Example 0.14 (Strictly convex with a singular Hessian)**.** Part 2 is sufficient but not necessary. $f(x) = x^4$ has

$$\begin{aligned} f''(0) &= 12 \cdot 0^2 \\ &= 0, \end{aligned}$$

R

```r
ch_x1_ch <- seq(-1, 1, length.out = 101)
ch_x2_ch <- seq(-3, 3, length.out = 101)
ch_grid_ch <- tidyr::expand_grid(x1 = ch_x1_ch, x2 = ch_x2_ch)
ch_u_ch <- 2 * ch_grid_ch$x1 + ch_grid_ch$x2
ch_grid_ch$z <- exp(ch_u_ch) - ch_grid_ch$x1
ch_line_x1_ch <- c(-1, 1)
ch_line_x2_ch <- c(2, -2)
ch_line_ch <- tibble::tibble(x1 = ch_line_x1_ch, x2 = ch_line_x2_ch)
ch_aes_ch <- ggplot2::aes(x1, x2)
ch_z_aes_ch <- ggplot2::aes(z = z)
ggplot2::ggplot(ch_grid_ch, ch_aes_ch) +
  ggplot2::geom_contour(ch_z_aes_ch, colour = "#0072B2", bins = 12) +
  ggplot2::geom_line(
    data = ch_line_ch, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::labs(x = "x1", y = "x2")
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np
```



```python
ch_x1_ax = np.linspace(-1, 1, 101)
ch_x2_ax = np.linspace(-3, 3, 101)
ch_xx_ax, ch_yy_ax = np.meshgrid(ch_x1_ax, ch_x2_ax)
ch_z_ax = np.exp(2 * ch_xx_ax + ch_yy_ax) - ch_xx_ax
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.contour(ch_xx_ax, ch_yy_ax, ch_z_ax, levels=12, colors="#0072B2")
ax.plot([-1, 1], [2, -2], "--", color="#D55E00")
```

R

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Python

```
import matplotlib.pyplot as plt
import numpy as np

ch_x1_3d, ch_x2_3d = np.meshgrid(np.linspace(-1, 1, 101),
                                 np.linspace(-3, 3, 101))
ch_z_3d = np.exp(2 * ch_x1_3d + ch_x2_3d) - ch_x1_3d
ch_line_x1_3d = np.linspace(-1, 1, 21)
ch_line_x2_3d = -2 * ch_line_x1_3d
ch_line_z_3d = np.exp(2 * ch_line_x1_3d + ch_line_x2_3d) - ch_line_x1_3d
fig = plt.figure(figsize=(6, 4.5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(ch_x1_3d, ch_x2_3d, ch_z_3d, color="#0072B2", alpha=0.6)
ax.plot(ch_line_x1_3d, ch_line_x2_3d, ch_line_z_3d, "--", color="#D55E00",
        label="f is linear")
ax.set_xlabel("x1")
ax.set_ylabel("x2")
ax.set_zlabel("f(x)")
ax.legend(loc="upper right")
plt.show();
```

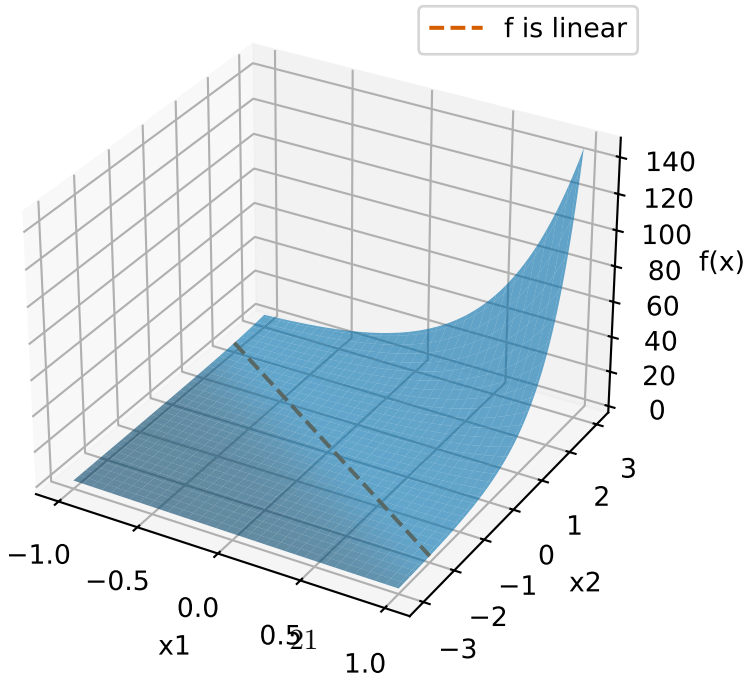


Figure 10: The graph of the third function in Example 0.13, the surface whose level curves are in Figure 9. Along the dashed line through the origin in the direction $(1, -2)^\top$ the surface is a straight line, so the function is linear there. The function is therefore

so $\mathbf{H}_f(0) = [0]$ is singular and not positive definite. Yet f is strictly convex, by the strict form of Equation 1: for $x \neq y$,

$$\begin{aligned} x^4 - y^4 - 4y^3(x-y) &= (x-y)(x^3 + x^2y + xy^2 + y^3) - 4y^3(x-y) && \text{(factor } x^4 - y^4\text{)} \\ &= (x-y)(x^3 + x^2y + xy^2 - 3y^3) && \text{(collect the } y^3 \text{ terms)} \\ &= (x-y)^2(x^2 + 2xy + 3y^2) && ((x-y)(x^2+2xy+3y^2) = x^3 + x^2y + x \\ &= (x-y)^2((x+y)^2 + 2y^2), && \text{(complete the square)} \end{aligned}$$

which is positive: $(x-y)^2 > 0$, and $(x+y)^2 + 2y^2 = 0$ would need $y = 0$ and then

$$\begin{aligned} x &= 0 \\ &= y. \end{aligned}$$

Theorem 0.7 (A convex function has a positive semidefinite Hessian)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$. If f is convex, then the Hessian $\mathbf{H}_f(\underset{\sim}{x})$ is positive semidefinite for every $\underset{\sim}{x} \in \mathbb{R}^p$.*

Proof

Proof. Fix $\underset{\sim}{x}, \underset{\sim}{v} \in \mathbb{R}^p$. Let $h > 0$. By Theorem 6 in Critical Points and Optimality Conditions applied with $\underset{\sim}{z} = \underset{\sim}{x}$ and $\underset{\sim}{h} = h\underset{\sim}{v}$, and again with $\underset{\sim}{h} = -h\underset{\sim}{v}$, there are $\tau_+, \tau_- \in (0,1)$ with

$$f(\underset{\sim}{x} + h\underset{\sim}{v}) = f(\underset{\sim}{x}) + h\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})\right)^{\top} \underset{\sim}{v} + \frac{h^2}{2}\, \underset{\sim}{v}^{\top} \mathbf{H}_f(\underset{\sim}{w}_+)\, \underset{\sim}{v} \quad \text{(Theorem 6 in Critical Points and Optimality Condi}$$

$$f(\underset{\sim}{x} - h\underset{\sim}{v}) = f(\underset{\sim}{x}) - h\left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})\right)^{\top} \underset{\sim}{v} + \frac{h^2}{2}\, \underset{\sim}{v}^{\top} \mathbf{H}_f(\underset{\sim}{w}_-)\, \underset{\sim}{v}, \quad \text{(Theorem 6 in Critical Points and Optimality Condi}$$

where $\underset{\sim}{w}_\pm = \underset{\sim}{x} \pm \tau_\pm h\underset{\sim}{v}$. Adding the two equations cancels the first-order summands:

$$f(\underset{\sim}{x} + h\underset{\sim}{v}) + f(\underset{\sim}{x} - h\underset{\sim}{v}) - 2f(\underset{\sim}{x}) = \frac{h^2}{2}\, \underset{\sim}{v}^{\top}\big(\mathbf{H}_f(\underset{\sim}{w}_+) + \mathbf{H}_f(\underset{\sim}{w}_-)\big)\underset{\sim}{v}.$$

Since f is convex and $\underset{\sim}{x} = \frac{1}{2}(\underset{\sim}{x} + h\underset{\sim}{v}) + \frac{1}{2}(\underset{\sim}{x} - h\underset{\sim}{v})$,

$$f(\underset{\sim}{x}) \leq \frac{1}{2} f(\underset{\sim}{x} + h\underset{\sim}{v}) + \frac{1}{2} f(\underset{\sim}{x} - h\underset{\sim}{v}),$$

so the left side of the previous display is nonnegative. Dividing by $h^2/2$ gives

$$\underset{\sim}{v}^{\top}\big(\mathbf{H}_f(\underset{\sim}{w}_+) + \mathbf{H}_f(\underset{\sim}{w}_-)\big)\underset{\sim}{v} \geq 0 \qquad \text{for every } h > 0.$$

As $h \to 0$, both $\underset{\sim}{w}_+$ and $\underset{\sim}{w}_-$ approach $\underset{\sim}{x}$, and the entries of $\mathbf{H}_f$ are continuous, so the left side approaches $2\,\underset{\sim}{v}^\top \mathbf{H}_f(\underset{\sim}{x})\,\underset{\sim}{v}$. Therefore $\underset{\sim}{v}^\top \mathbf{H}_f(\underset{\sim}{x})\,\underset{\sim}{v} \geq 0$ for every $\underset{\sim}{v}$. □

Corollary 0.1 (Convex if and only if the Hessian is positive semidefinite)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$. Then f is convex if and only if the Hessian $\mathbf{H}_f(\underset{\sim}{x})$ is positive semidefinite for every $\underset{\sim}{x} \in \mathbb{R}^p$.*

Proof

Proof. Part 1 of Theorem 0.6 proves that a positive semidefinite Hessian at every point makes f convex, and Theorem 0.7 proves that a convex f has a positive semidefinite Hessian at every point. □

Theorem 0.8 (Second-derivative test for concavity)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$.*

1. *f is concave if and only if $-\mathbf{H}_f(\underset{\sim}{x})$ is positive semidefinite for every $\underset{\sim}{x} \in \mathbb{R}^p$.*
2. *If $-\mathbf{H}_f(\underset{\sim}{x})$ is positive definite for every $\underset{\sim}{x} \in \mathbb{R}^p$, then f is strictly concave.*

Proof

Proof. Its negative $-f$ has the second partial derivatives $-\frac{\partial}{\partial x_j}\frac{\partial}{\partial x_k} f$, which are continuous on $\mathbb{R}^p$, so its Hessian is $\mathbf{H}_{-f}(\underset{\sim}{x}) = -\mathbf{H}_f(\underset{\sim}{x})$.

1. By Definition 0.4, f is concave if and only if $-f$ is convex. By Corollary 0.1 applied to $-f$, that holds if and only if $-\mathbf{H}_f(\underset{\sim}{x})$ is positive semidefinite for every $\underset{\sim}{x}$.
2. Part 2 of Theorem 0.6, applied to $-f$, makes $-f$ strictly convex, so f is strictly concave by Definition 0.5.

□

Example 0.15 (A strictly concave quadratic function)**.** Let $f(\underset{\sim}{x}) = 1 - x_1^2 - x_2^2$ for $\underset{\sim}{x} = (x_1, x_2)^\top$. Its Hessian is $\mathbf{H}_f(\underset{\sim}{x}) = -2\mathbf{I}_2$ at every $\underset{\sim}{x}$, so $-\mathbf{H}_f(\underset{\sim}{x}) = 2\mathbf{I}_2$. For every $\underset{\sim}{v} \neq \underset{\sim}{0}$,

$$\begin{aligned} \underset{\sim}{v}^\top (2\mathbf{I}_2)\underset{\sim}{v} &= 2\underset{\sim}{v}^\top \underset{\sim}{v} && (\mathbf{I}_2\underset{\sim}{v} = \underset{\sim}{v}) \\ &= 2\|\underset{\sim}{v}\|^2 && (\underset{\sim}{v}^\top \underset{\sim}{v} = \|\underset{\sim}{v}\|^2) \\ &> 0, && (\underset{\sim}{v} \neq \underset{\sim}{0}) \end{aligned}$$

so $-\mathbf{H}_f(\underset{\sim}{x})$ is positive definite, and f is strictly concave by part 2 of Theorem 0.8. Figure 11 shows the dome that its graph forms.

R

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Python

```
import matplotlib.pyplot as plt
import numpy as np

cc_x1_py, cc_x2_py = np.meshgrid(np.linspace(-1.5, 1.5, 41),
                                 np.linspace(-1.5, 1.5, 41))
cc_dome_py = 1 - cc_x1_py**2 - cc_x2_py**2
fig = plt.figure(figsize=(6, 4.5))
ax = fig.add_subplot(projection="3d")
ax.plot_surface(cc_x1_py, cc_x2_py, cc_dome_py, color="#0072B2", alpha=0.7)
ax.set_xlabel("x1")
ax.set_ylabel("x2")
ax.set_zlabel("f(x)")
plt.show();
```

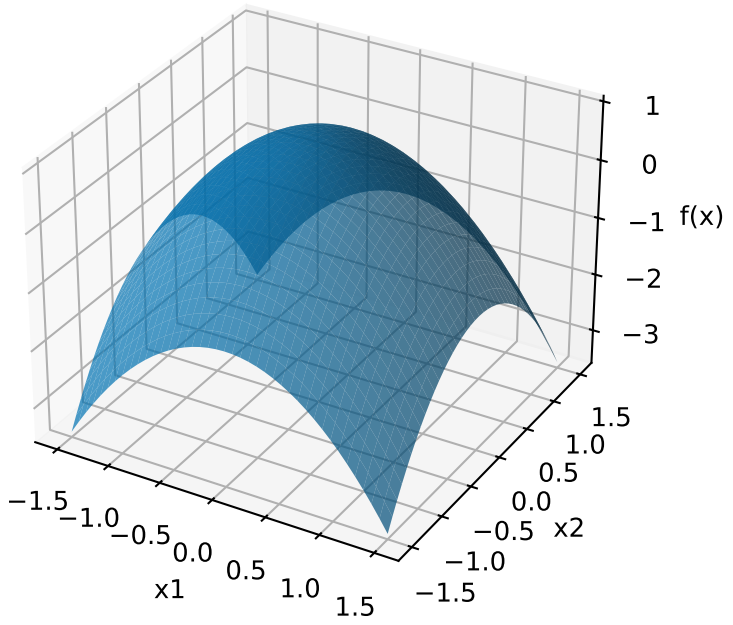


Figure 11: The graph of the strictly concave function $f(\underset{\sim}{x}) = 1 - x_1^2 - x_2^2$ of Example 0.15 is a dome with its highest point at $\underset{\sim}{0}$.

Corollary 0.2 (Stationary points of a convex function are global minimizers)**.** *If f is as in Theorem 0.5 and convex, then every stationary point $\underset{\sim}{x}^*$ of f (Definition 4 in Critical Points and Optimality Conditions) is a global minimizer.*

Proof

Proof. For every $\underset{\sim}{x}$,

$$\begin{aligned} f(\underset{\sim}{x}) &\geq f(\underset{\sim}{x}^*) + \left(\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^*)\right)^\top (\underset{\sim}{x} - \underset{\sim}{x}^*) && \text{(Equation 1 with } \underset{\sim}{y} = \underset{\sim}{x}^*) \\ &= f(\underset{\sim}{x}^*) + \underset{\sim}{0}^\top (\underset{\sim}{x} - \underset{\sim}{x}^*) && (\underset{\sim}{x}^* \text{ is stationary}) \\ &= f(\underset{\sim}{x}^*). && (\underset{\sim}{0} \cdot \underset{\sim}{v} = 0) \end{aligned}$$

□

Example 0.16 (A convex function, and a non-convex one)**.**

- $f(\underset{\sim}{x}) = x_1^2 + x_2^2 - 2x_1$ is convex by Theorem 0.6, since its Hessian is $2\mathbf{I}_2$ (Example 16 in Critical Points and Optimality Conditions), so its stationary point $(1, 0)^\top$ (Example 8 in Critical Points and Optimality Conditions) is a global minimizer, as completing the square showed in Example 9 in Critical Points and Optimality Conditions.
- Without convexity, a stationary point need not be a minimizer at all: $x_1^2 - x_2^2$ has the stationary point $\underset{\sim}{0}$, which is not a minimizer (Example 9 in Critical Points and Optimality Conditions).
- The corollary says nothing about whether a minimizer exists: $e^{2x_1 + x_2} - x_1$ is convex (Example 0.13) but has no stationary point, and so no local minimizer (Example 9 in Critical Points and Optimality Conditions).

Definition 0.10 (Convex optimization problem, convex problem)**.** A minimization problem whose objective function is convex and whose feasible set is a convex set is a **convex optimization problem** (or **convex problem**) [5].

Example 0.17 (Minimizing over a line, and over a circle)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$. Its Hessian is $2\mathbf{I}_2$ at every $\underset{\sim}{x}$, which is positive semidefinite, so f is convex by part 1 of Theorem 0.6.

- Minimizing f subject to $3x_1 + 4x_2 = 5$, as in the example of a minimization problem, is a convex problem. Its feasible set $L = \{\underset{\sim}{x} \in \mathbb{R}^2 : 3x_1 + 4x_2 = 5\}$ is a convex set: for $\underset{\sim}{x}, \underset{\sim}{y} \in L$ and $t \in [0, 1]$,

$$
\begin{aligned}
3(tx_1 + (1-t)y_1) + 4(tx_2 + (1-t)y_2) &= t(3x_1 + 4x_2) + (1-t)(3y_1 + 4y_2) && \text{(distribute and regroup)} \\
&= 5t + 5(1-t) && (\underset{\sim}{x}, \underset{\sim}{y} \in L) \\
&= 5, && \text{(arithmetic)}
\end{aligned}
$$

so $t\underset{\sim}{x} + (1-t)\underset{\sim}{y} \in L$.

- Minimizing the same f subject to $x_1^2 + x_2^2 = 1$ is not a convex problem, because its feasible set, the unit circle C, is not a convex set: $(1,0)^\top$ and $(-1,0)^\top$ are in C, but with $t = \frac{1}{2}$ the point $\underset{\sim}{0}$ is not, since $0^2 + 0^2 = 0 \neq 1$.

Figure 12 shows both feasible sets.

R

```
cop_x1_fc <- seq(-1.5, 2.5, length.out = 2)
cop_x2_fc <- (5 - 3 * cop_x1_fc) / 4
cop_line_fc <- tibble::tibble(x1 = cop_x1_fc, x2 = cop_x2_fc)
cop_angle_fc <- seq(0, 2 * pi, length.out = 201)
cop_cos_fc <- cos(cop_angle_fc)
cop_sin_fc <- sin(cop_angle_fc)
cop_circle_fc <- tibble::tibble(x1 = cop_cos_fc, x2 = cop_sin_fc)
cop_line_pts_x1_fc <- c(-1, 1)
cop_line_pts_x2_fc <- c(2, 0.5)
cop_line_pts_fc <- tibble::tibble(
  x1 = cop_line_pts_x1_fc,
  x2 = cop_line_pts_x2_fc
)
cop_circle_pts_x1_fc <- c(-1, 1)
```

Theorem 0.9 (The exponential function lies above its tangent line at zero)**.** *For every real x,*

$$\exp\{x\} \geq 1 + x,$$

with equality exactly when $x = 0$.

Proof

Proof. Let $g(x) = \exp\{x\} - 1 - x$. The exponential function has second derivative $\exp\{x\} > 0$ at every x, so it is convex by Theorem 0.6. The function g differs from it by a linear function, so g is convex too. The derivative of g is $\frac{dg(x)}{dx} = \exp\{x\} - 1$, which is 0 at $x = 0$. By Corollary 0.2, 0 is a global minimizer of g, and $g(0) = 0$, so $g(x) \geq 0$ for every x.
Suppose $g(x) = 0$ for some x. Then x is also a global minimizer of g, and so a local minimizer. Fermat's theorem gives $\frac{dg(x)}{dx} = \exp\{x\} - 1 = 0$. Then $\exp\{x\} = 1 = \exp\{0\}$, which gives $x = 0$ because the exponential function is injective. □

Corollary 0.3 (The natural logarithm lies below its tangent line at one)**.** *For every $t > 0$,*

$$\log\{t\} \leq t - 1,$$

with equality exactly when $t = 1$.

Proof

Proof. Put $x = \log\{t\}$, so that $\exp\{x\} = t$ by the inverse relation of the logarithm and the exponential. By Theorem 0.9, $t = \exp\{x\} \geq 1+x = 1+\log\{t\}$, which is $\log\{t\} \leq t-1$. Equality holds exactly when $x = 0$, that is, when $\log\{t\} = 0$, that is, when $t = 1$. □

Example 0.18 (Two values of the bounds)**.**
At $x = 1$, $\exp\{1\} = \mathrm{e} \approx 2.718$, which is at least $1 + 1 = 2$. At $t = 2$, $\log\{2\} \approx 0.693$, which is at most $2 - 1 = 1$. Neither is an equality, because $x \neq 0$ and $t \neq 1$.

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[3] Wikipedia contributors, "Epigraph (mathematics) — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Epigraph_(mathematics)

[4] Wikipedia contributors, "Tangent space — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Tangent_space

[5] Wikipedia contributors, "Convex optimization — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Convex_optimization